

Daniel Yan

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EDUCATION

University of Toronto

BASc in Mechanical Engineering + PEY Co-op

- GPA: 3.49 — Dean's Honour List

Toronto, ON

Sep 2024 – Expected Apr 2028

SKILLS

Technical Skills: SolidWorks, Ansys, CATIA, Abaqus, CATIA Composites Design, 3DEXPERIENCE, Python

Tools: MS Office, Slack, Git, Gantt Chart

Manufacturing: Composites Layup, 3D Printing, Composite Molding, Stackup Design

Certificates: CSWA, G2 Driver's License, George Brown Machining (lathe, mill)

Languages: Mandarin (Fluent), English (Native)

EXPERIENCE

Structural & Fabrication Lead

Toronto, ON

Blue Sky Solar Racing - UofT Design Team

Apr 2025 - Current

- Optimized the bottom aerobody ply schedule using CATIA Composites Design; validated race regulation compliance via Ansys FEA to achieve a **56% component weight reduction** and a **10% decrease** in vehicle mass
- Designed a roll cage in CATIA, leveraging topology optimization over **17 iterations** to meet race regulations, achieving **40% mass reduction** while maintaining an average minimum **safety factor of 1.2** under **10** regulatory load cases
- Spearheaded the adoption of **CATIA Composites Design** to optimize ply layup precision and streamline CAD-to-ANSYS FEA workflows, eliminating material waste from manual cutting errors by **63%**; facilitated knowledge transfer to **20+** members via a self-authored **50+ page** technical manual to standardize tool proficiency
- Led end-to-end development of a driver ergonomics test platform to optimize cockpit layout, ingress/egress biomechanics, and safety system integration; translated multi-driver measurements into chassis design inputs, reducing egress time by **5 seconds** under race conditions
- Designed a ventilated paint booth for component finishing, ensuring all specifications were met for Environmental, Health, and Safety (**EHS**) **compliance**; managing a Bill of Materials achieving a **53%** cost saving over retail models
- Refined solar car chassis performance by designing a comprehensive composite layup sequence in 3DEXPERIENCE, providing the foundational data for structural analysis and future carbon fiber refinements
- Directed composite wet layup and infusion for car components; enforced EHS standards and specialized equipment SOPs to ensure 100% structural reliability and team safety during high-pressure build cycles

Team Leader

Greater Toronto Area, ON

Nutters Jazz Band

Feb 2022 - Aug 2024

- Directed the creation of a jazz band by managing cross-school logistics, communication, and rehearsal operations, culminating in a **17-member** ensemble from **8** schools, gathered from professional music events
- Coordinated rehearsals, repertoire, and managed **10+** paid performances/charity events across the GTA
- Established teamwork, leadership, organization, and management skills through positive feedback from clients

PROJECTS

Linear Algebra Visualization App | Python, Flask, Manim, PyWebView

Dec 2024 - Current

- Built an interactive desktop application featuring custom user input capabilities to **visualize linear algebra concepts** (matrix transformations, change of basis) with real-time rendering
- Packaged the app into a portable installer with PyInstaller, bundling FFmpeg and MiKTeX for full offline usability across computers
- Showcased to a University of Toronto professor and adopted by students at **10+** universities across Canada to enhance learning through dynamic visualization

Badminton Shuttlecock Launcher | SolidWorks, ESP32, C++

May 2025 - Current

- Designed and engineered a **fully-automatic 2-DOF** launching system in SolidWorks with pan/tilt capabilities to cover the entire badminton court, featuring adjustable trajectory height up to **4 meters** and maximum shuttlecock **speed of 180 km/h** for realistic **single-player** training scenarios
- Integrated electromechanical systems including high-speed brushless motors with ESCs, servo motors, ESP32, and a LiPo battery; implemented PWM control and fault-tolerant wiring to ensure consistent performance
- Developed **custom C++ control software** with HTML-based Bluetooth interface for iPhone connectivity, with fully customizable training sequences for various drill types